

ES di prova

$$P \rightarrow Q, \neg P \models \neg Q$$

$$\downarrow$$
$$P \rightarrow Q \models \neg Q \rightarrow \neg P$$

$$\forall v. \llbracket P \rightarrow Q \rrbracket_v = 1 \Rightarrow \llbracket \neg Q \rightarrow \neg P \rrbracket_v = 1$$

$$\llbracket P \rightarrow Q \rrbracket_v = 1 \Leftrightarrow \llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 1 \Rightarrow \llbracket \neg Q \rightarrow \neg P \rrbracket_v = 1$$

$$\Leftrightarrow \llbracket \neg Q \rrbracket_v = 1 \text{ OR } \llbracket \neg P \rrbracket_v = 0 \rightarrow \text{VERO}$$

ES di prova

$$P \rightarrow Q, \neg Q \models \neg P$$

$$\forall v. \llbracket (P \rightarrow Q), \neg Q \rrbracket_v = 1 \Rightarrow \llbracket \neg P \rrbracket_v = 1$$

$$\llbracket (P \rightarrow Q), \neg Q \rrbracket_v = 1 \Leftrightarrow \llbracket P \rightarrow Q \rrbracket_v = 1 \text{ AND } \llbracket \neg Q \rrbracket_v = 1 \Leftrightarrow$$

$$(\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 1) \text{ AND } \llbracket Q \rrbracket_v = 0 \Rightarrow \llbracket \neg P \rrbracket_v = 1 \Leftrightarrow$$

$$\llbracket P \rrbracket_v = 0 \rightarrow \text{FALSO}$$

ES: 1

$$P \rightarrow Q, Q \models P$$

$$\forall v. \llbracket (P \rightarrow Q), Q \rrbracket_v = 1 \Rightarrow \llbracket P \rrbracket_v = 1$$

$$\llbracket (P \rightarrow Q), Q \rrbracket_v = 1 \Leftrightarrow \llbracket P \rightarrow Q \rrbracket_v = 1 \text{ AND } \llbracket Q \rrbracket_v = 1 \Leftrightarrow$$

$$(\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 1) \text{ AND } \llbracket Q \rrbracket_v = 1 \Rightarrow \llbracket P \rrbracket_v = 1 \rightarrow \text{FALSA}$$

ES 2:

$$P \rightarrow Q, \neg P \models \neg Q$$

$$\forall v. \llbracket (P \rightarrow Q), \neg P \rrbracket_v = 1 \Rightarrow \llbracket \neg Q \rrbracket_v = 1$$

$$\llbracket (P \rightarrow Q), \neg P \rrbracket_v = 1 \Leftrightarrow \llbracket P \rightarrow Q \rrbracket_v = 1 \text{ AND } \llbracket \neg P \rrbracket_v = 1 \Leftrightarrow$$

$$(\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 1) \text{ AND } \llbracket P \rrbracket_v = 0 \Rightarrow \llbracket Q \rrbracket_v = 0 \rightarrow \text{FALSO}$$

ES 3:

$$(P \wedge Q) \rightarrow R, \neg R \models \neg P$$

$$\forall v. \llbracket ((P \wedge Q) \rightarrow R), \neg R \rrbracket_v = 1 \Rightarrow \llbracket \neg P \rrbracket_v = 1$$

$$\llbracket ((P \wedge Q) \rightarrow R), \neg R \rrbracket_v = 1 \Leftrightarrow \llbracket (P \wedge Q) \rightarrow R \rrbracket_v = 1 \text{ AND } \llbracket \neg R \rrbracket_v = 1$$

$$\Leftrightarrow ((\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 0) \text{ OR } \llbracket R \rrbracket_v = 1) \text{ AND } \llbracket R \rrbracket_v = 0 \Rightarrow$$

$$\llbracket \neg P \rrbracket_v = 1 \Leftrightarrow \llbracket P \rrbracket_v = 0 \rightarrow \text{VERA}$$

ES 4:

$$P \rightarrow R \models (P \vee Q) \rightarrow R$$

$$\forall v. \llbracket P \rightarrow R \rrbracket_v = 1 \Rightarrow \llbracket (P \vee Q) \rightarrow R \rrbracket_v = 1$$

$$\begin{aligned} \llbracket P \rightarrow R \rrbracket_v = 1 &\Leftrightarrow \llbracket P \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1 \Rightarrow \llbracket (P \vee Q) \rightarrow R \rrbracket_v = 1 \\ &\Leftrightarrow (\llbracket P \vee Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \Leftrightarrow ((\llbracket P \rrbracket_v = 0 \text{ AND } \llbracket Q \rrbracket_v = 0) \\ &\text{OR } \llbracket R \rrbracket_v = 1) \rightarrow \text{FALSA} \end{aligned}$$

ES 5:

$$((P \rightarrow Q) \rightarrow P) \models P$$

$$\forall v. \llbracket ((P \rightarrow Q) \rightarrow P) \rrbracket_v = 1 \Rightarrow \llbracket P \rrbracket_v = 1$$

$$\begin{aligned} \llbracket ((P \rightarrow Q) \rightarrow P) \rrbracket_v = 1 &\Leftrightarrow \llbracket (P \rightarrow Q) \rrbracket_v = 0 \text{ OR } \llbracket P \rrbracket_v = 1 \Leftrightarrow \\ &(\llbracket P \rrbracket_v = 1 \text{ AND } \llbracket Q \rrbracket_v = 0) \text{ OR } \llbracket P \rrbracket_v = 1 \Rightarrow \llbracket P \rrbracket_v = 1 \rightarrow \text{VERA} \end{aligned}$$

ES 6:

$$(P \rightarrow R) \vee (Q \rightarrow R) \models (P \vee Q) \rightarrow R$$

$$\forall v. \llbracket (P \rightarrow R) \vee (Q \rightarrow R) \rrbracket_v = 1 \Rightarrow \llbracket (P \vee Q) \rightarrow R \rrbracket_v = 1$$

$$\begin{aligned} \llbracket (P \rightarrow R) \vee (Q \rightarrow R) \rrbracket_v = 1 &\Leftrightarrow (\llbracket P \rightarrow R \rrbracket_v = 1 \text{ OR } \llbracket Q \rightarrow R \rrbracket_v = 1) \Leftrightarrow \\ &((\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \text{ OR } (\llbracket Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1)) \Rightarrow \\ \llbracket (P \vee Q) \rightarrow R \rrbracket_v = 1 &\Leftrightarrow (\llbracket P \vee Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \Leftrightarrow \\ &((\llbracket P \rrbracket_v = 0 \text{ AND } \llbracket Q \rrbracket_v = 0) \text{ OR } \llbracket R \rrbracket_v = 1) \rightarrow \text{FALSA} \end{aligned}$$

ES 7:

$$(P \rightarrow R) \wedge (Q \rightarrow R) \models (P \vee Q) \rightarrow R$$

$$\forall v. \llbracket (P \rightarrow R) \wedge (Q \rightarrow R) \rrbracket_v = 1 \Rightarrow \llbracket (P \vee Q) \rightarrow R \rrbracket_v = 1$$

$$\begin{aligned} \llbracket (P \rightarrow R) \wedge (Q \rightarrow R) \rrbracket_v = 1 &\Leftrightarrow \llbracket P \rightarrow R \rrbracket_v = 1 \text{ AND } \llbracket Q \rightarrow R \rrbracket_v = 1 \Leftrightarrow \\ &(\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \text{ AND } (\llbracket Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \Rightarrow \\ \llbracket (P \vee Q) \rightarrow R \rrbracket_v = 1 &\Leftrightarrow \llbracket P \vee Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1 \Leftrightarrow \\ &(\llbracket P \rrbracket_v = 0 \text{ AND } \llbracket Q \rrbracket_v = 0) \text{ OR } \llbracket R \rrbracket_v = 1 \rightarrow \text{VERA} \end{aligned}$$

ES 8:

$$(P \wedge Q) \rightarrow R \models P \rightarrow (Q \rightarrow R)$$

$$\forall v. \llbracket (P \wedge Q) \rightarrow R \rrbracket_v = 1 \Rightarrow \llbracket P \rightarrow (Q \rightarrow R) \rrbracket_v = 1$$

$$\begin{aligned} \llbracket (P \wedge Q) \rightarrow R \rrbracket_v = 1 &\Leftrightarrow (\llbracket P \wedge Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \Leftrightarrow \\ &(\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 0) \text{ OR } \llbracket R \rrbracket_v = 1 \Rightarrow \\ \llbracket P \rightarrow (Q \rightarrow R) \rrbracket_v = 1 &\Leftrightarrow \llbracket P \rrbracket_v = 0 \text{ OR } (\llbracket Q \rrbracket_v = 0 \text{ OR } \llbracket R \rrbracket_v = 1) \\ &\rightarrow \text{VERA} \end{aligned}$$

ES 9:

$$(P \rightarrow Q) \vee (R \rightarrow S) \models (P \vee R) \rightarrow (Q \vee S)$$

$$\forall v. \llbracket (P \rightarrow Q) \vee (R \rightarrow S) \rrbracket_v = 1 \Rightarrow \llbracket (P \vee R) \rightarrow (Q \vee S) \rrbracket_v = 1$$

$$\llbracket (P \rightarrow Q) \vee (R \rightarrow S) \rrbracket_v = 1 \Leftrightarrow \llbracket (P \rightarrow Q) \rrbracket_v = 1 \text{ OR } \llbracket (R \rightarrow S) \rrbracket_v = 1 \Leftrightarrow$$

$$(\llbracket P \rrbracket_v = 0 \text{ OR } \llbracket Q \rrbracket_v = 1) \text{ OR } (\llbracket R \rrbracket_v = 0 \text{ OR } \llbracket S \rrbracket_v = 1) \Rightarrow$$

$$\llbracket (P \vee R) \rightarrow (Q \vee S) \rrbracket_v = 1 \Leftrightarrow (\llbracket P \vee R \rrbracket_v = 0 \text{ OR } \llbracket Q \vee S \rrbracket_v = 1) \Leftrightarrow$$

$$(\llbracket P \rrbracket_v = 0 \text{ AND } \llbracket R \rrbracket_v = 0) \text{ OR } (\llbracket Q \rrbracket_v = 1 \text{ OR } \llbracket S \rrbracket_v = 1)$$

$\rightarrow$ FALSE